

Reviewer 3

This revised manuscript evaluates the contribution of multi-source imagery (i.e., radar, optical, and elevation data) for tropical peatland mapping. Furthermore, different classification scenarios were investigated using Random Forest (RF) algorithm and assessment of land cover classes to image features was done.

This manuscript needs a major revision before it is considered for potential publication in this journal. My major concerns are listed as follows:

- In Abstract term „Random forest-recursive feature elimination“ is mentioned. Recursive feature elimination (RFE) is an embedded type of feature selection algorithm, but as mentioned in Abstract, reader might think it is a built-in Random Forest method. Also, LN 322 refers recursive feature elimination as REF, and later in manuscript RFE is used.

RESPONSE: Thanks for your suggestion. The random forest term has been removed as recommended.

- Table 2 and further in manuscript, class Natural forest is mentioned, whereas later the authors mention and refer to the Mixed forest. This needs to be adjusted

RESPONSE: The term has been standardized to natural forest throughout the manuscript.

- Table 3, S2+ - which texture features were used in this classification scenario?

RESPONSE: Texture was computed from the standard deviation of the NDVI. This detail has been added to the table.

- LN 332: please describe in detail about Gini index and mention which another measure for feature importance scores exists

RESPONSE: Thanks for the comment. Fuller description of Gini index and other measures has been provided in section 2.5.

- Eq 14: F-score equation is presented as f1 and different abbreviation is used later

RESPONSE: This has been standardized to F-score throughout the manuscript.

- Section 2.3. mentions training and testing data, and Section 2.5. addressed as „... and evaluation“ the reader does not receive the information about validation protocol (sampling design, how the split between train and validation data is made (and the purpose of using polygon and point data for classification)).

Please take a look at these two papers:

"Key issues in rigorous accuracy assessment of land cover products" by Stehman & Foody (2019).

"Good practices for estimating area and assessing accuracy of land change" by Olofsson & al (2014).

RESPONSE: Thanks for your comment. The text in Sections 2.3 and 2.5 has been modified to ensure consistency. Data from polygon extents were used for model building (i.e. feature selection and training) while the point-based data served as an independent sample for accuracy assessment. More details on the training/testing split have been added to the text under Section 2.5.

- Figure 3: caption on y-axis – what does it mean „no of correct classification“?

RESPONSE: The y-axis has been edited to remove this confusion.

- Table 5 and 6: a detailed preview of the results has been made, but it is pretty difficult to derive some conclusions from them. I suggest to put OA, UA and PA in the range from 0-100, and in Table 6 perhaps use bold and italic values to emphasise best UA/PA value of each class

RESPONSE: Thanks for your comment. These tables have been altered as suggested, with data values expressed as %s and the best OA, UA and PA in bold.

- LN 464: I do not think that the extent and distribution is shown on Figure 6

RESPONSE: Thanks for your comment. Figure 6 has been changed to figure 8 in the text.

- Figure 7 – in what coordinate system are expressed x and y axis? Ranges over longitude/latitude cover a wide area according to the range of each axis

RESPONSE: Thanks for your comment. Erroneous coordinates have been removed.